



When Fair Enough Is Not Enough

Allocation Policies and Long-Term Benefit Gaps

PAPER 1

Zezulka & Genin

From the fair distribution of predictions to the fair distribution of social goods: Evaluating the impact of fair machine learning on long-term unemployment.

PAPER 2

Scher et al.

Modelling the long-term fairness dynamics of data-driven targeted help on job seekers.

PAPER 3

Xu et al.

Adapting static fairness to sequential decision-making: Bias mitigation strategies towards equal long-term benefit rate.

MOTIVATION

Real algorithm  Real decision  Real person

Can a model that passes a fairness check today still create unfair outcomes tomorrow?

- Public agencies increasingly use ML scores to allocate support.
- These decisions affect access to training, loans, admissions, and other social goods.
- The problem is not only whether the score is fair, but what happens after deployment.



Austria, 2019: the AMS algorithm

The Austrian Public Employment Service began scoring every unemployed person into three groups: high, medium, or low prospects of finding a job.

Training support was sent only to the medium group, because it looked efficient on paper.

Women, older workers, people with care duties, and other vulnerable groups were systematically marked low-prospects,.

Caretake feature → flag for women.

Static fairness is not enough



- Static metrics evaluate the model before deployment.
- Deployed models change decisions and future outcomes.
- Long-term fairness depends on the policy, the environment, and time.

Zeulka & Genin: fairness as a policy problem

- Fairness should be evaluated on social goods, not only scores.
- After deployment, the score changes the allocation decision.
- The same score can support opposite policies: help high-risk people or exclude them.
- Prospective fairness asks what outcome the policy will produce.

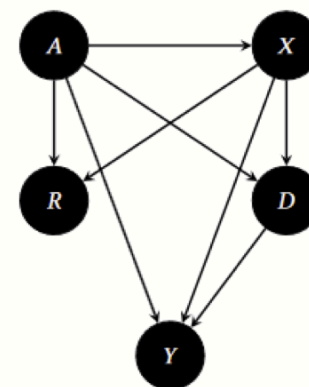
A = sensitive attribute

X = features

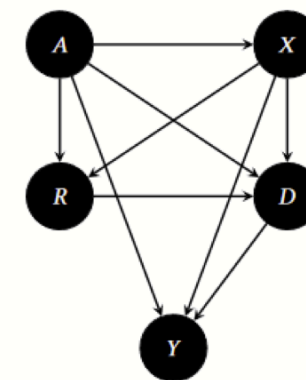
R = risk score

D = decision

Y = outcome



(a) Causal structure G_{pre} before deploying an algorithmically informed policy.



(b) Causal structure G_{post} after deploying an algorithmically informed policy.

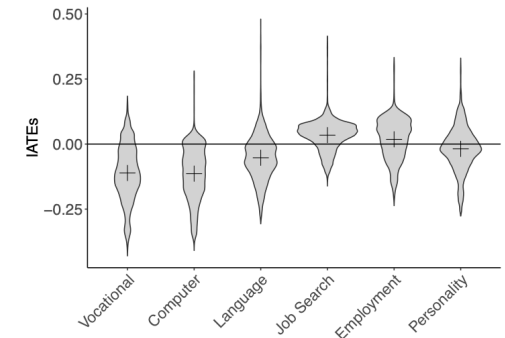
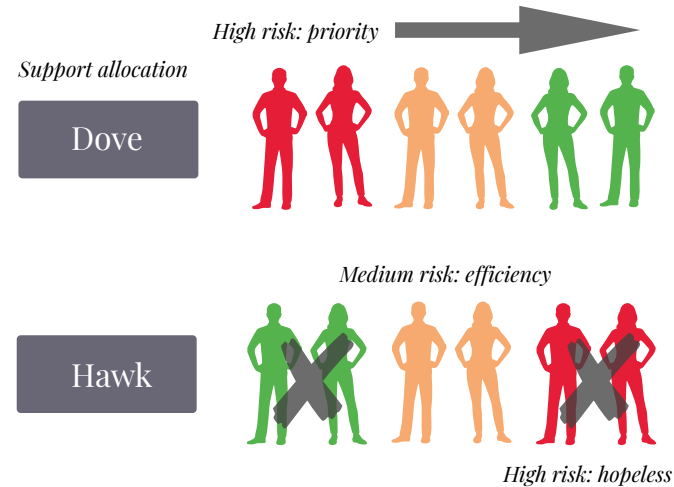
Zeulka & Genin Figure 1: causal graph structure before and after deploying an algorithmically informed policy.

After deployment the risk score becomes part of the causal system

Fair risk scores can worsen downstream outcomes

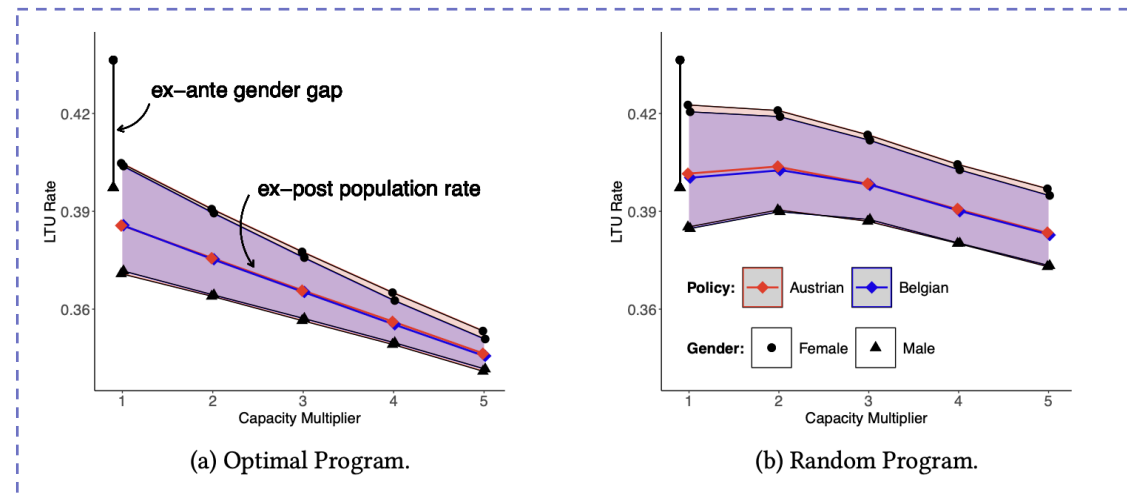
- Case study: Swiss long-term unemployment.
- Compare Dove (Belgian) vs Hawk (Austrian) allocation.
- Fairness-constrained scores can increase the gender gap.
- Optimal program assignment improves outcomes. Some programs are more effective.

low-risk: below the 30th percentile
medium-risk: between the 30th and 70th percentiles
high-risk: above the 70th percentile.



(a) Individualized Average Treatment Effects.

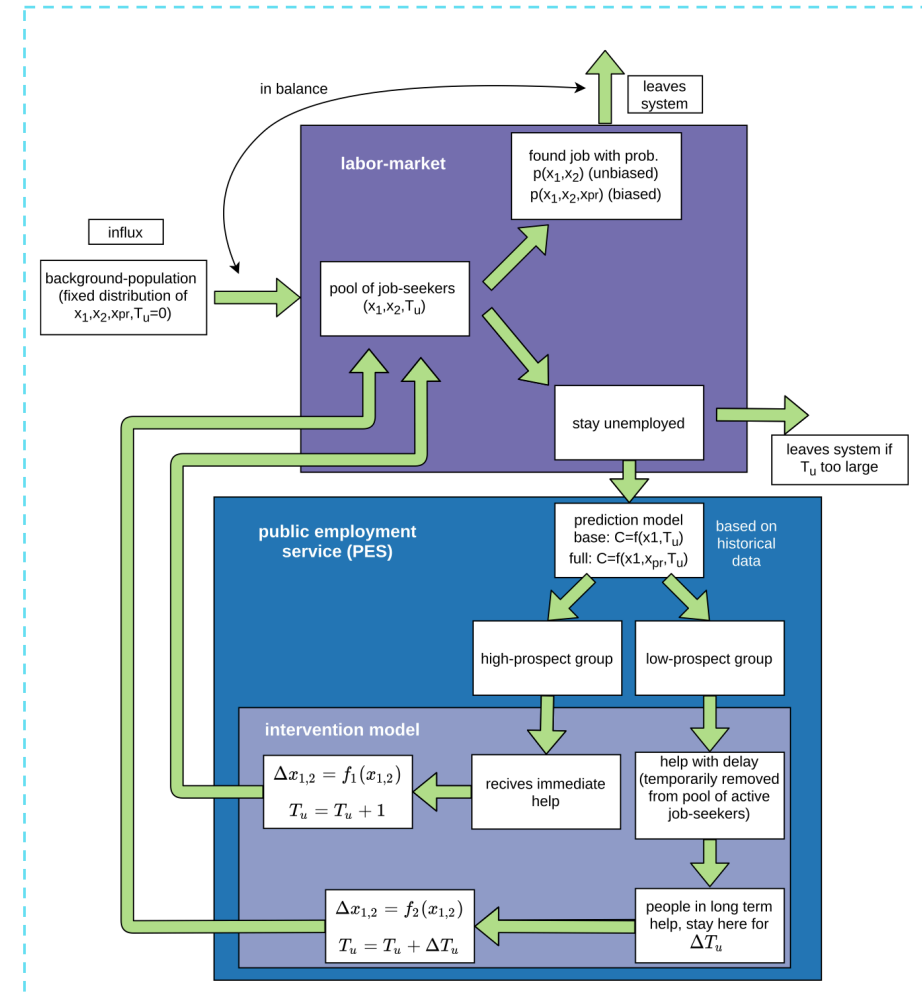
Zezulka & Genin Figure 3: Negative treatment effects imply a lower risk of becoming long-term unemployed



Zezulka & Genin Figure 5: Overall LTU (Long Term Unemployment) and gender gap as a function of program capacity.

Scher et al.: the environment is part of the fairness system

- A public agency does not act on a static pool of job seekers.
- Job seekers enter, leave, receive help, and change skills.
- Fairness must be evaluated inside a dynamic labor market.



Scher et al. Figure 2: outline of the labor market and PES (Public Employment Service) model.

Simulating targeted help in a labor market



Setup:

- Synthetic job seekers with two skills and one protected attribute.
- Public Employment Service observes only partial skills.
- Two prediction models:
 - Base model: does not use protected attribute.
 - Full model: uses protected attribute.
- Support changes skills: labor market decides who finds a job.
- Biased labor market and unbiased labor market

Question:

Does using group information reduce inequality or create unfair individual treatment?

Group improvement can hide individual harm

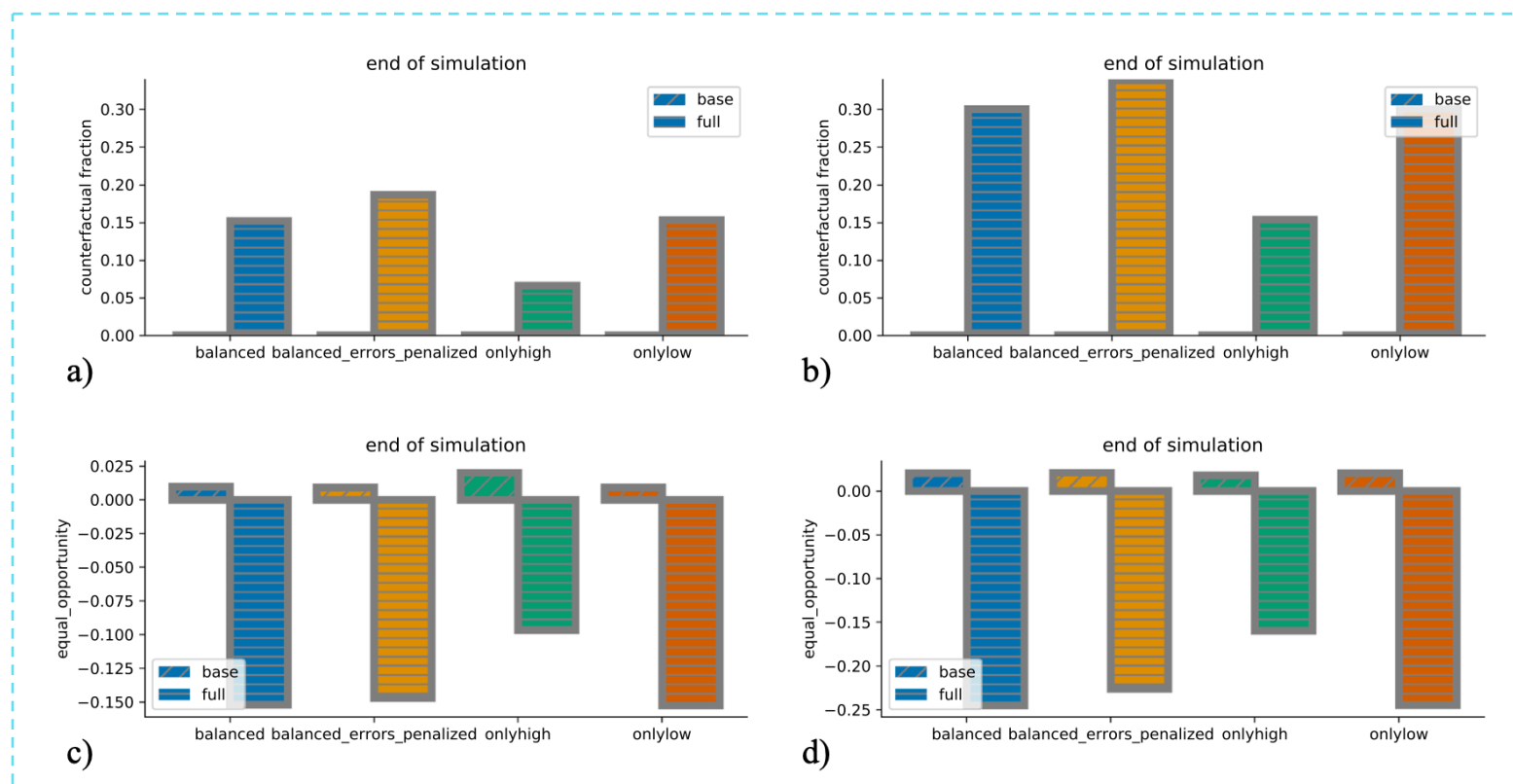
- The full model can use protected group information to target more help.
- This may reduce group-level skill gaps.
- But it also creates counterfactual cases: people labeled low-prospect only because of group membership.
- The trade-off is only visible when simulated over time. Main trade-off: group disparity reduction vs individual misclassification

Balanced: Everyone receives the same type of help, no targeting.

Only low: Worse labor-market prospects

Only high: high-prospect receive help

Balanced errors penalized: Everyone receives help, but the help is less effective when the model misclassifies the person.



Scher et al. Figure 6 : counterfactual fraction at the end. (a,c) biased market (b,d) unbiased market

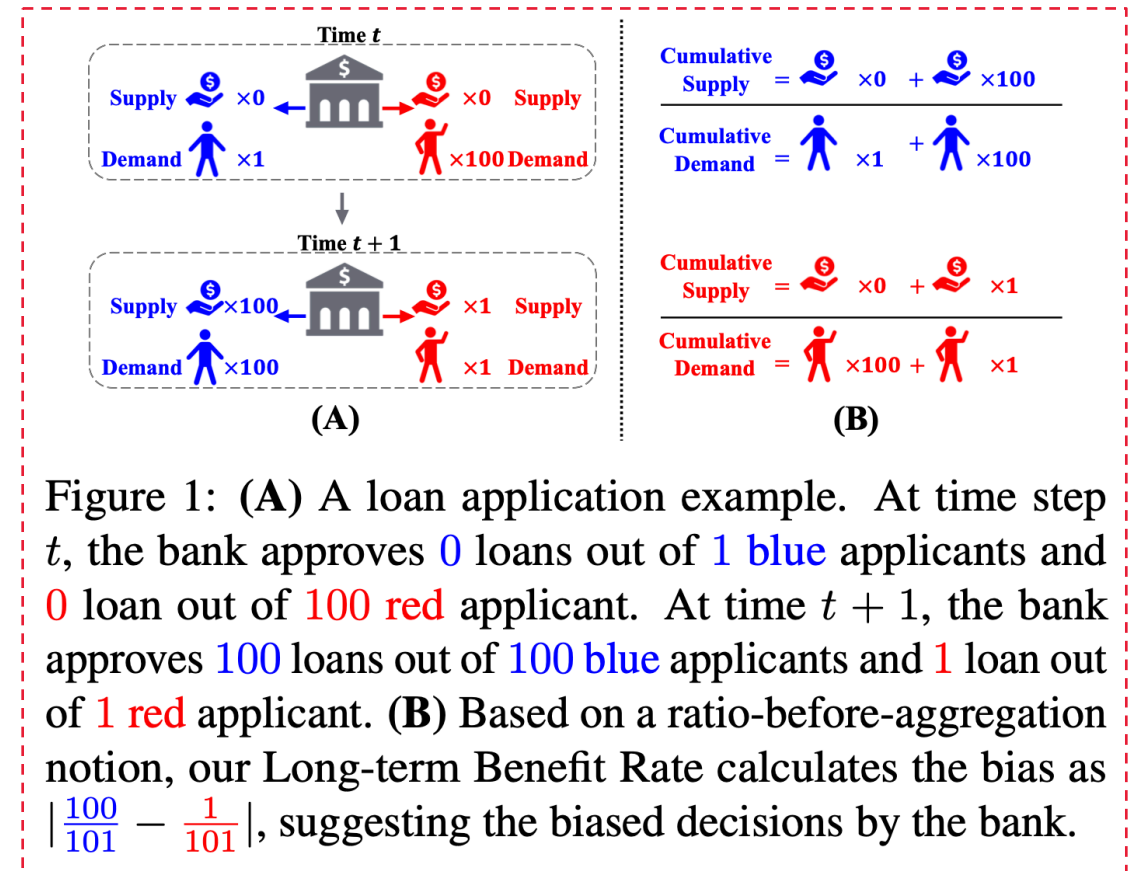
Xu et al.: fairness depends on how we aggregate over time

- Step-wise fairness can hide long-term bias.
- ELBERT (Equal Long-term Benefit Rate) measures cumulative benefit over cumulative demand.
- Main idea: aggregate first, divide after.

Markov Decision Process (MDP): framework for sequential decisions. At each step, the system has a state, the public agency takes an action, receives a reward, and moves to a new state.

$$\text{Long-term benefit rate} = \frac{\text{total benefit received}}{\text{total need over time}}$$

aggregate, then divide



Xu et al. Figure 1: loan application example.

ELBERT reduces long-term bias while preserving utility

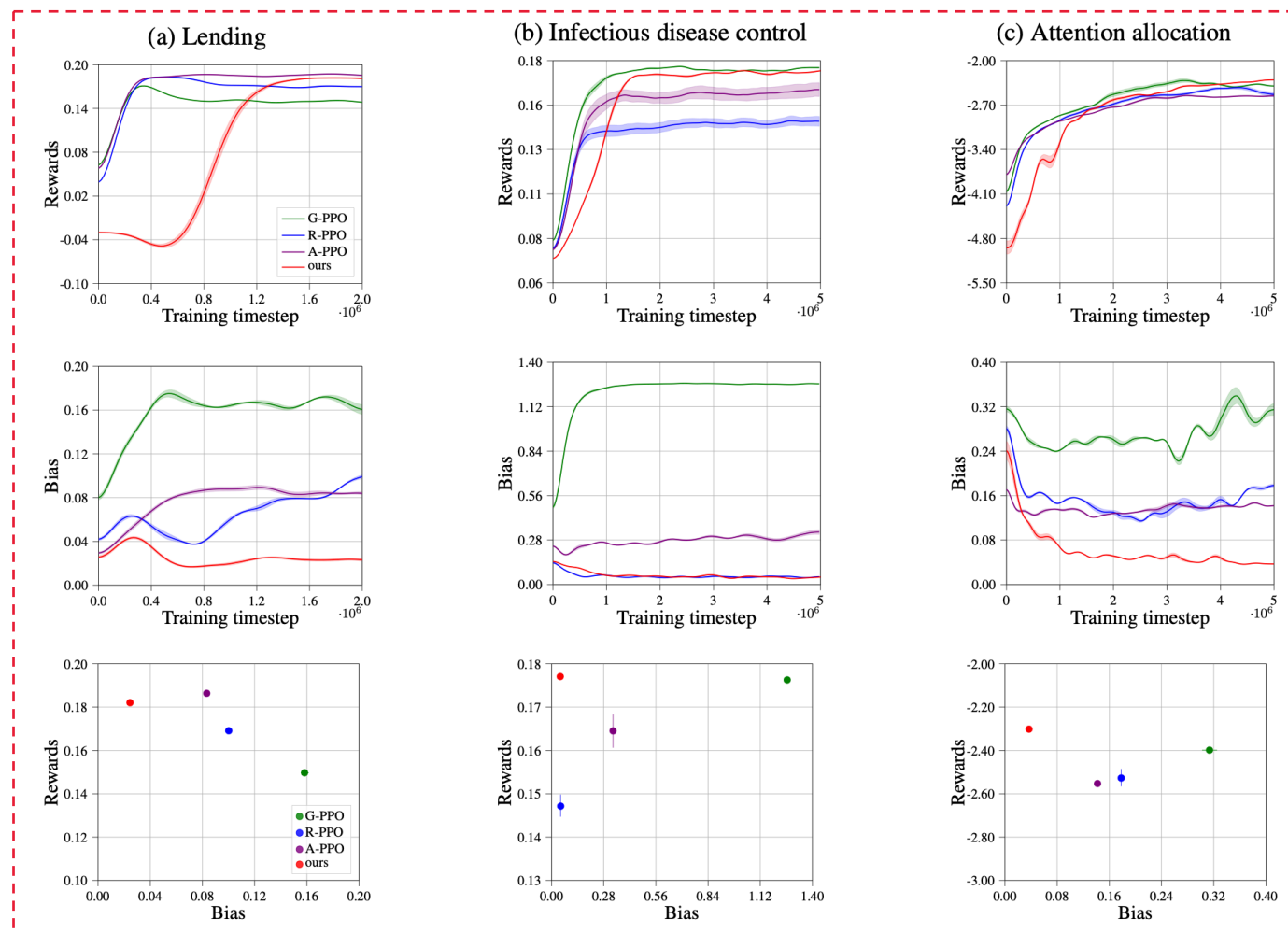
- Supply/Demand MDP adds fairness signals to an MDP.
- ELBERT-PO (Policy Optimization) optimizes reward with a long-term fairness penalty.
- Experiments compare reward and bias across environments.
- Main result: lower long-term bias with competitive reward.

Greedy PPO : maximizes reward

Reward-only fairness constrained PPO: fairness penalty directly into the reward.

Advantage-regularized PPO: fairer next-step decisions

*PPO is a reinforcement learning algorithm that improves a policy step by step



Xu et al. Figure 4: reward and bias of ELBERT-PO vs baselines.

Three complementary lenses for long-term fairness

POLICY

Zezulka & Genin

What outcome does the score produce?

ENVIRONMENT

Scher et al.

How does the system evolve after intervention?

TIME AGGREGATION

Xu et al.

How should fairness accumulate overtime?

Long-term fairness requires policy + environment + time aggregation.

Experiment

Paper	Idea
Zezulka & Genin	Policy
Scher et al.	Environment
Xu et al.	Metric over time & penalty

Sensitive group: SEX

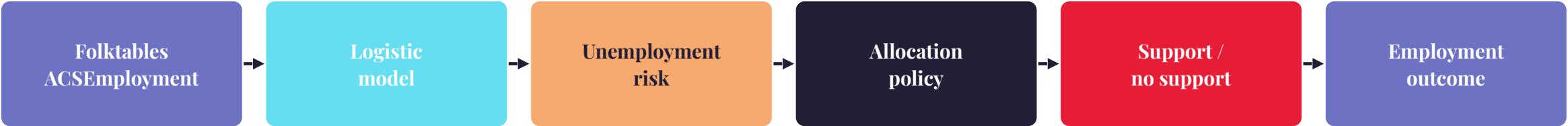
Baseline pattern: Male group has higher unemployment risk but women higher need rate.

Average Unemployment risk

Female 30% vs Male 33%

Need rate.

Female 36% vs Male 27%



Goal: test how static allocation rules behave once support affects future employment probabilities

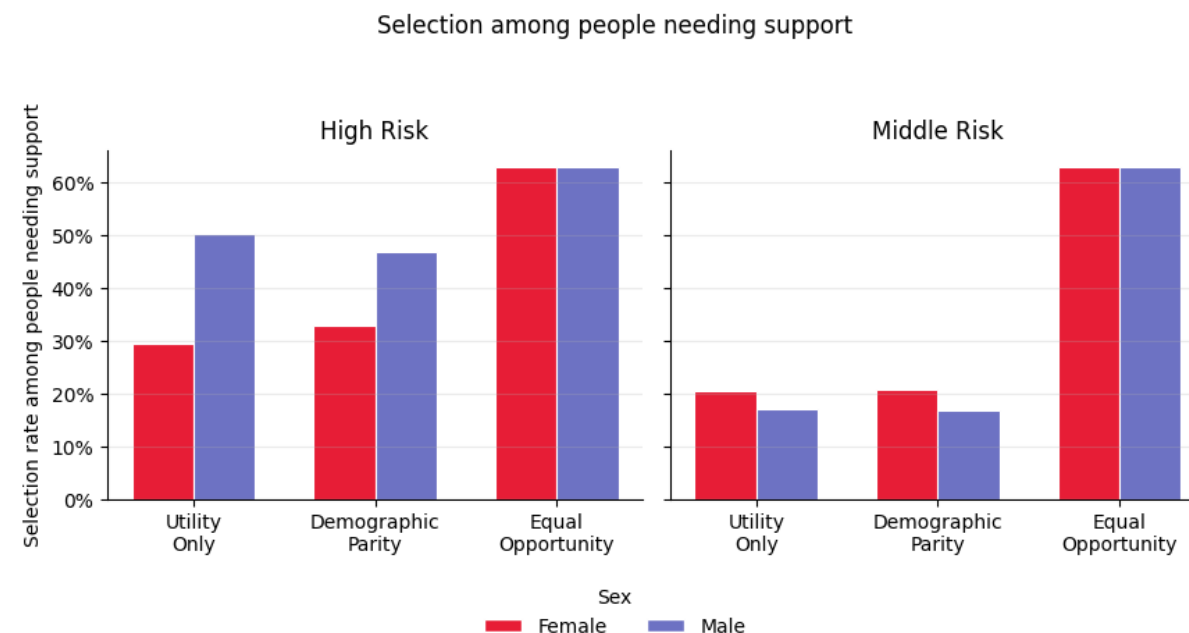
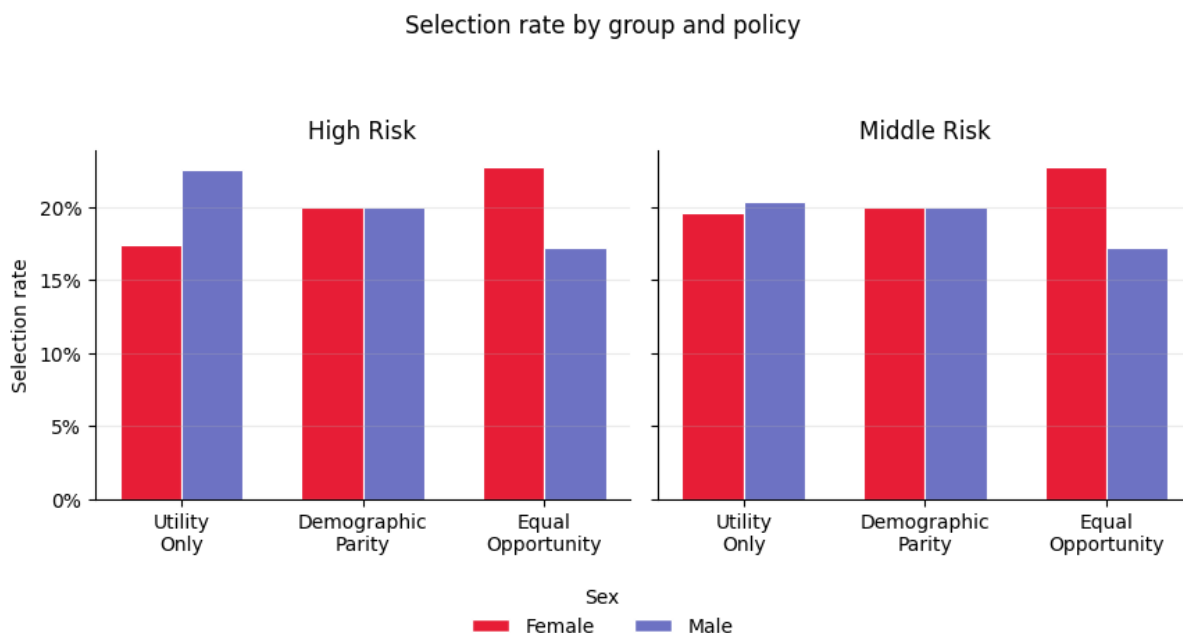
Static allocation policies: constraints and capacity



- Utility-only: prioritize people with highest predicted unemployment risk.
- Demographic parity: equalize overall selection rates by group.
- Equal opportunity: equalize selection among people who need support.
- Different budgets: increase support capacity, changing how scarce help is.
- Targeting variants: who receives support first

Policy choice	What changes?
Utility-only	Prioritizes highest predicted risk
Demographic Parity (DP)	Equalizes selection rates by group
Equal Opportunity (EO)	Equalizes selection among high-need people
Expanded budget	Budget as policy choice
High vs middle risk	Changes which risk group is treated as the priority population

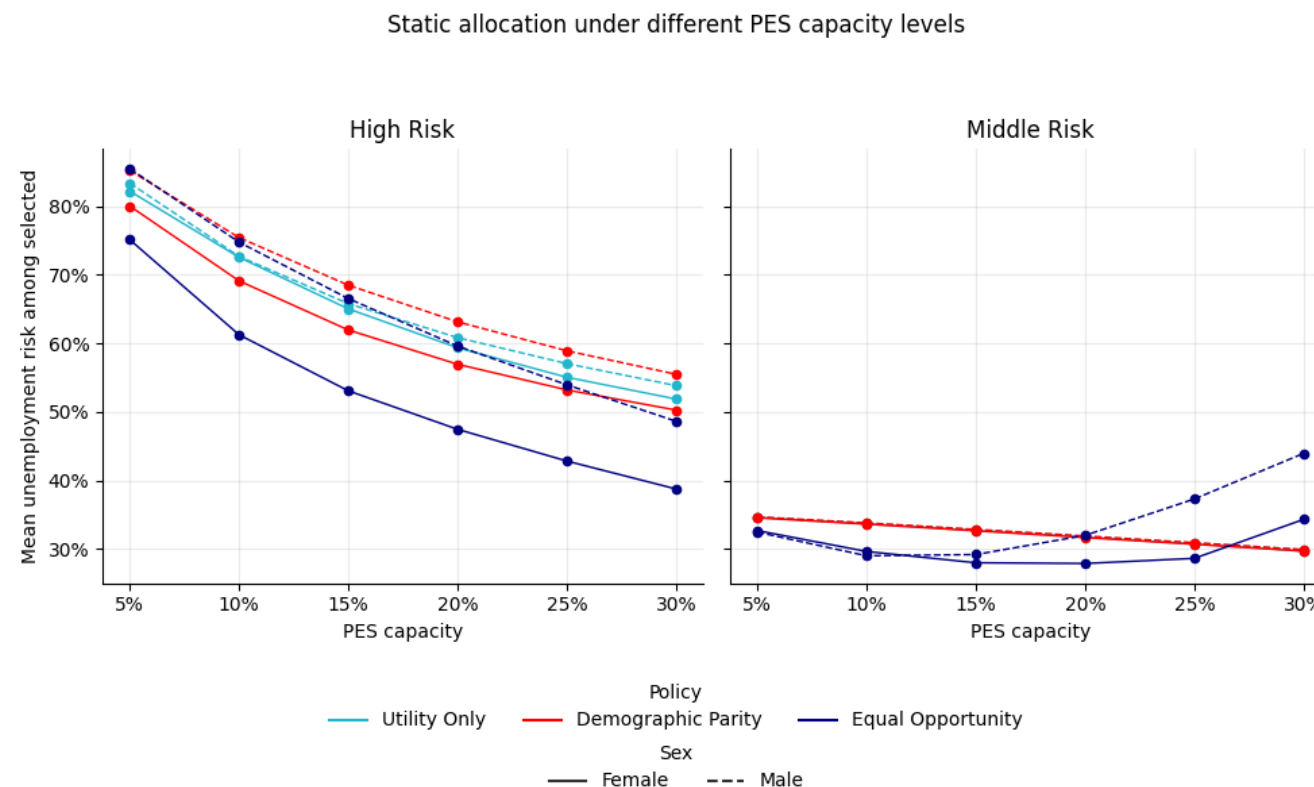
Different static rules solve different fairness problems



- Male group has higher unemployment risk but women higher need rate.
- Utility-only HR selects more men individuals because it prioritizes measured risk.
- DP reduces the selection gap by equalizing group selection rates.
- EO equalizes selection among people needing support.
- MR targeting lowers the selected-risk profile and becomes less need-focused.

Budget as a policy choice

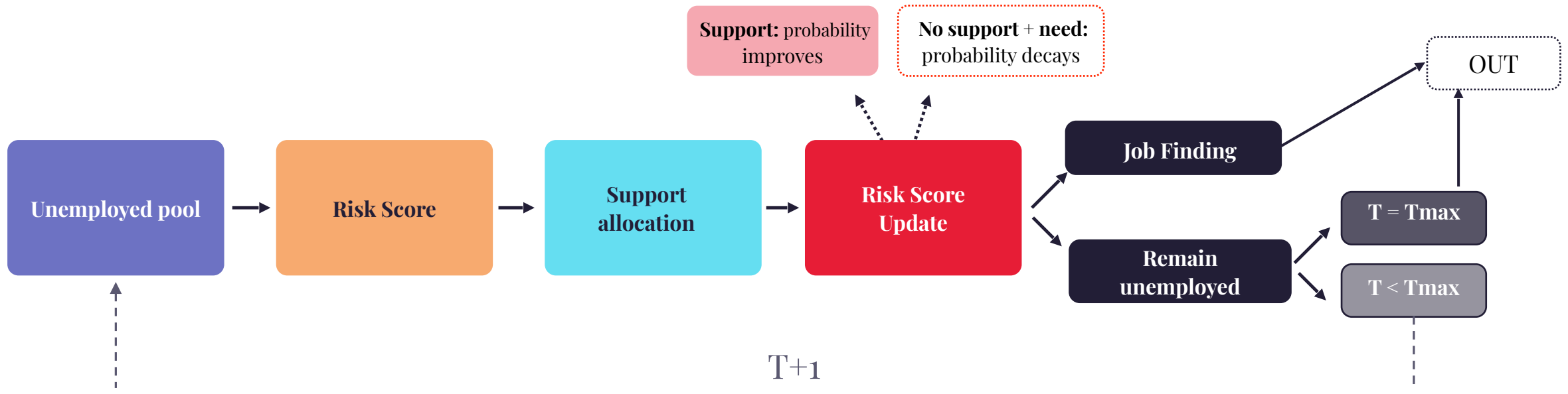
- Increasing capacity expands who can receive support.
- Under HR targeting, larger budgets move beyond the highest-risk cases.
- Under MR targeting, selected risk is lower and less tied to need.
- Capacity reduces scarcity, but priority rules still determine who is reached first.



More capacity expands access, but targeting determines who is reached first.

From one allocation to repeated consequences

- Each round starts with a pool of unemployed people.
- A policy selects who receives support under a limited budget.
- Supported people receive a positive update to employment probability (25% stronger program effect).
- Unsupported high-need people can experience probability decay.
- People who find jobs leave the pool, timed-out cases also exit, the pool is refilled.
- Metrics are averaged across rounds and seeds



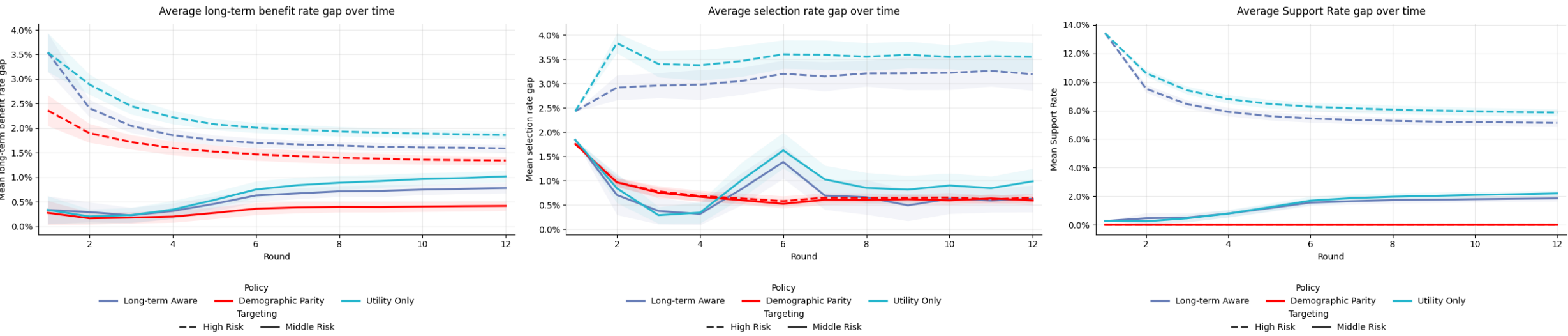
Allocation fairness vs outcome fairness

Metric	What it checks
Selection rate gap	Who receives support in each round (selected group x at time T/ total group X at time T)
Support rate gap	Of all the times each group was in the pool, what share of those opportunities received support?
Benefit rate gap	Who accumulates benefit relative to need (close to ELBERT idea: cumulative benefit received / cumulative need over time)
Job-finding gap among need	Outcome gap among high-need people
Timeout rate	Non-employment exits after max unemployment duration (T=6)

- Selection gaps are static diagnostics.
- Benefit gaps are cumulative long-term diagnostics.

MAIN RESULT

Static fairness does not fully determine cumulative benefit



Policy variant	Benefit gap	Selection gap	Job finding	Timeout rate
DP, middle-risk	0.004	0.006	0.522	0.019
DP, high-risk	0.013	0.006	0.548	0.010
Long-term-aware, middle-risk	0.008	0.006	0.521	0.019
Long-term-aware, high-risk	0.016	0.032	0.549	0.009
Utility-only, middle-risk	0.010	0.010	0.522	0.019
Utility-only, high-risk	0.019	0.035	0.549	0.009

- Demographic Parity produces the smallest benefit gaps under both target groups.
- HR targeting has higher job-finding among the high-need pool.
- MR targeting selects easier cases but leaves more high-need people to time out (% no-employment exit)**
- Low selection gap alone is not enough: target population changes the long-term outcome.

TAKEAWAY

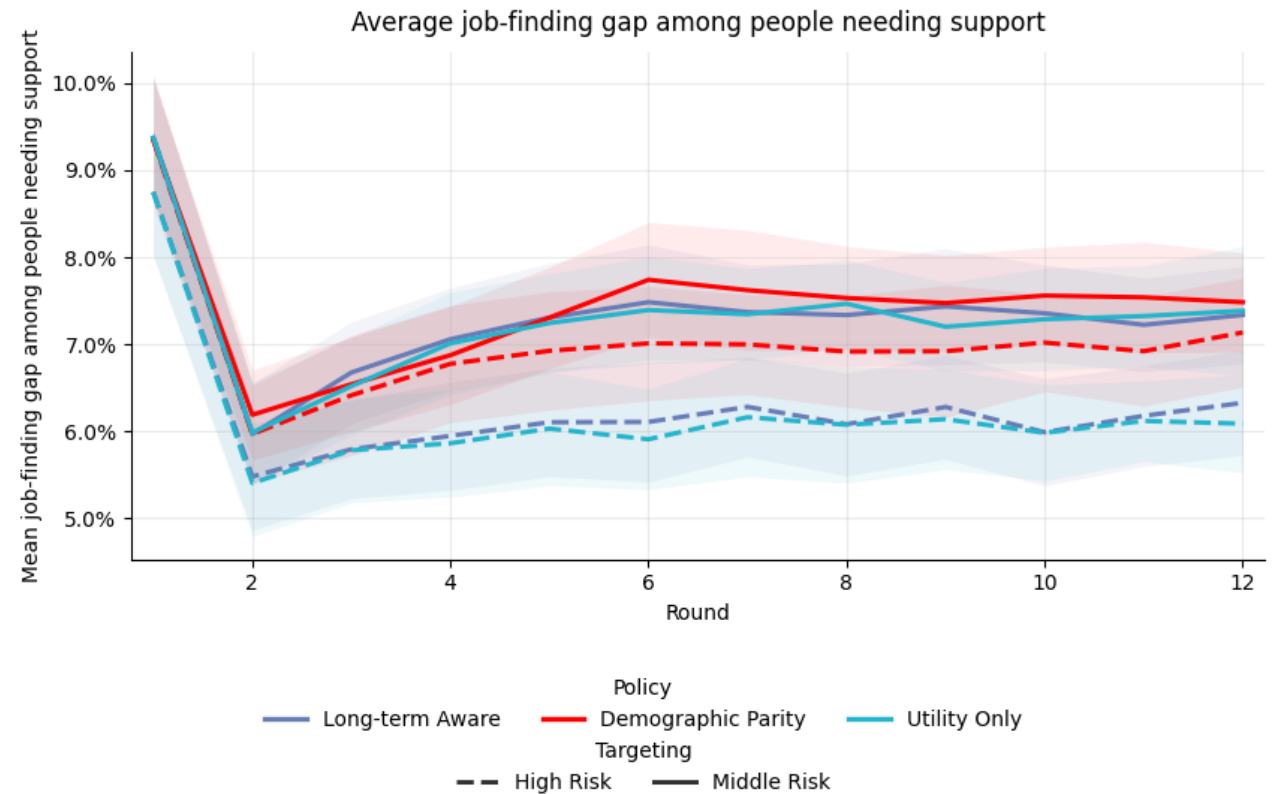
Fair enough is not enough

- **Zezulka & Genin:** evaluate the policy outcome, not only the score.
- **Scher et al.:** model the environment because interventions change the system.
- **Xu et al.:** aggregate fairness over time, not only step by step.
- **My experiment:** equalizing selection is not enough because groups differ in baseline risk and need

Fairness should be evaluated where benefits and harms accumulate:
over people, policies, and time.

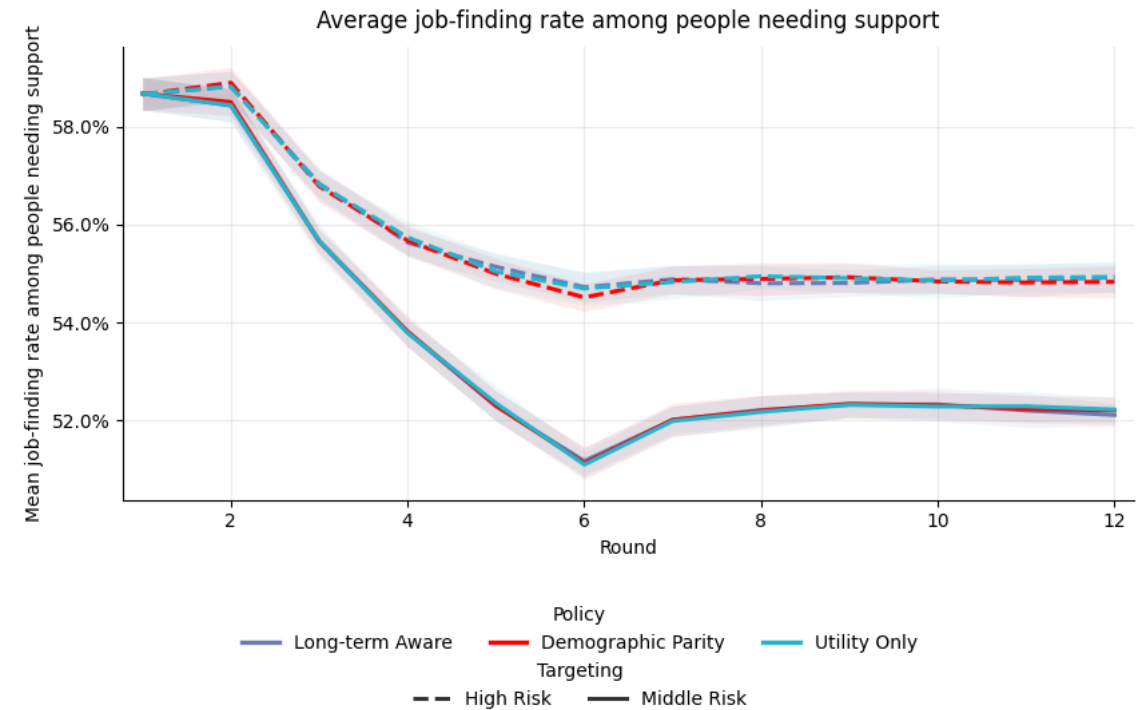
Job-finding gap among people needing support

- Outcome-based check among the high-need population.
- Complements allocation gaps.
- Shows whether support translates into similar job-finding outcomes.



Job-finding rate among people needing support

- Utility measure for the high-need population.
- Separates effectiveness from fairness.
- Checks whether gap reduction comes at the cost of lower outcomes.



Final-round metrics by group

Policy Variant	Race	Selection Rate	Selection Among Need	Job-finding rate	Benefit Rate	Support Rate Cumulative	Avg Risk Among Need
Long-term (HR)	Female	0.097	0.166	0.581	0.044	0.165	0.418
	Male	0.129	0.234	0.517	0.060	0.236	0.483
Long-term (MR)	Female	0.111	0.189	0.558	0.110	0.191	0.440
	Male	0.118	0.211	0.485	0.118	0.209	0.514
Static DP (HR)	Female	0.117	0.200	0.584	0.060	0.200	0.415
	Male	0.110	0.200	0.513	0.046	0.200	0.486
Static DP (MR)	Female	0.118	0.200	0.560	0.116	0.200	0.439
	Male	0.112	0.200	0.485	0.112	0.200	0.514
Utility Only (HR)	Female	0.095	0.163	0.580	0.043	0.161	0.418
	Male	0.131	0.237	0.519	0.062	0.240	0.482
Utility Only (MR)	Female	0.109	0.186	0.559	0.109	0.189	0.439
	Male	0.119	0.214	0.486	0.119	0.211	0.514

- Shows where each aggregate gap comes from.
- Separates support, benefit, employment, and job-finding.
- Useful for interpreting final-round differences by group.